



BOILEAU GUILLAUME

Chef de Projet Logiciel / Système | Spécifications, Exigences, Validation, Coordination Technique & Reporting

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EXPERIENCE

Software Engineer - System Integration, Hardware Interfaces & Production Follow-up

VEV Platform Services France

📅 2025 – 2026

📍 Valbonne, France

Context: Development and integration of software services for an EV charging CPMS platform connected to operational infrastructure and field equipment.

Objective: Support reliable delivery, production follow-up, software integration and operational data-flow continuity between platform services and charging station systems.

- **Software delivery:** Developed and maintained web and backend services using TypeScript, Angular and Node.js in an operational industrial environment.
- **System integration:** Contributed to the integration and maintenance of software services interacting with field equipment, charging stations and operational infrastructure.
- **Needs clarification:** Translated operational issues, data-flow constraints, equipment feedback and user needs into technical tasks and corrective actions.
- **Production follow-up:** Analysed service behaviour, data exchanges, communication issues and anomalies affecting operational continuity.
- **Verification / validation:** Checked data consistency, interface continuity and service behaviour in production-like or operational conditions.
- **Technical issue management:** Investigated incidents involving software behaviour, operational data exchanges and hardware-connected systems.
- **Coordination:** Worked with technical stakeholders to follow priorities, clarify issues and support iterative delivery.
- **Data exchange:** Implemented and monitored FTP-based operational data exchange services.
- **Field / equipment exposure:** Hands-on exposure to EV charging stations, test support and electrical-safety constraints in an industrial context.
- **Agile tooling:** Used Zenhub to support task tracking, backlog follow-up, iterative development and technical coordination.

Senior R&D Engineer - System Software, Validation, Project Coordination & Reporting

CNES, Laboratoire Lagrange, Observatoire de la Côte d'Azur

📅 2023 – 2025

📍 Nice, France

Context: Engineering activities for the LISA ESA/NASA space mission, involving complex software chains, model integration, simulation workflows, validation campaigns and international coordination.

Objective: Structure, coordinate and validate complex software/data workflows under strong consistency, traceability, performance and verification constraints.

- **Project coordination:** Coordinated technical activities involving contributors, research engineers, technicians and Master's thesis interns.
- **Specifications / cadrage:** Structured development priorities, validation logic, expected outputs and technical acceptance criteria for complex analysis workflows.
- **Solution implementation:** Developed CATpyLISA, a Python platform for large-scale model integration, comparison, validation and technical review. [GitLab: CATpyLISA](#)
- **V-cycle logic:** Supported activities aligned with specification, integration, verification, validation, traceability and technical evidence consolidation.
- **Verification / validation:** Defined comparison campaigns, consistency checks, acceptance logic and validation workflows for technical outputs.
- **Risk follow-up:** Identified deviations, inconsistencies, limiting factors and technical risks affecting result reliability or delivery quality.
- **Reporting:** Produced technical notes, validation summaries, analysis reports and review material for contributors and project stakeholders.
- **Documentation:** Used Confluence to support technical documentation, coordination notes, validation follow-up and knowledge sharing.
- **Stakeholder alignment:** Worked at the interface between scientific requirements, software development, data processing, modelling assumptions and mission-level objectives.
- **International environment:** Operated in a collaborative framework requiring technical English, structured communication and versioned workflows.

Data Scientist - Technical Studies, Performance Analysis & Project Reporting

Universiteit Antwerpen, Belgium

📅 2021 – 2023

📍 Antwerp, Belgium

Context: Data analysis and performance studies for precision instruments in a high-requirement international environment linked to gravitational-wave detector performance.

Objective: Identify, characterize and mitigate factors affecting system sensitivity using statistical, computational and validation-oriented methods.

- **Technical studies:** Analysed instrumental and environmental effects impacting system sensitivity, stability and measurement quality.
- **Analysis workflows:** Developed Python-based pipelines for noise source identification, uncertainty estimation and statistical modelling.

PROFILE FOR DAVIDSON

Software / systems project engineer with strong experience in requirements logic, technical coordination, validation, integration, reporting and delivery follow-up for complex systems. Background developed in international space-mission environments and reinforced by recent industrial software experience involving operational services, hardware/software interfaces and production follow-up.

For a **Chef de projet logiciel / système embarqué** role, I bring a project-oriented technical profile: writing and structuring specifications, tracing needs and constraints, coordinating contributors, following risks and deliverables, challenging technical answers, supporting verification / validation and communicating clearly with clients and stakeholders.

- Strong fit with software/system project follow-up: specifications, requirements logic, validation, technical reporting and stakeholder coordination.
- Experience coordinating multidisciplinary contributors across software, data, modelling, operational constraints and system-level objectives.
- Ability to translate technical needs, anomalies and constraints into structured actions, deliverables and follow-up.
- Strong analytical ability to investigate hard points, identify risks, challenge assumptions and converge toward pragmatic solutions.
- Practical experience with software integration, APIs, data flows, Linux environments, containers, Git-based workflows and operational monitoring.
- Industrial exposure to hardware/software interfaces through EV charging infrastructure, operational data exchanges and equipment-connected services.
- Comfortable working in English and in multidisciplinary environments involving technical and non-technical stakeholders.

FIT FOR EMBEDDED SOFTWARE / SYSTEMS

- Relevant background for system/software project phases: needs analysis, specification logic, architecture review support, implementation follow-up, verification, validation and delivery support.
- Strong experience producing technical documentation, validation reports, progress summaries and decision-support material.
- Experience coordinating contributors, clarifying priorities, identifying risks and supporting corrective actions.
- Practical understanding of software components: APIs, backend services, data flows, containers, Linux, Git/GitLab, CI/CD and operational monitoring.
- Experience at the interface between software behaviour, operational data, system constraints and hardware-connected infrastructure.
- Comfortable with collaborative tools including Git/GitLab, GitHub, Confluence, Zenhub and JIRA-style issue tracking.

LANGUAGES

English ● ● ● ● ●

Spanish ● ● ● ● ●

EDUCATION

PhD in Physics

Université Côte d'Azur

📅 2018 – 2021

📍 Nice, France

- **Performance analysis:** Identified contributors to performance degradation and supported prioritisation of mitigation directions.
- **Data-driven decisions:** Analysed measurement and simulation outputs to identify abnormal behaviours and dominant contributors.
- **Documentation:** Produced reproducible and traceable technical analyses for international engineering and research stakeholders.
- **Coordination:** Communicated complex technical results through structured reporting, presentations and collaborative review.

Junior R&D Engineer - Simulation, Software Workflows & Validation

ARTEMIS, Observatoire de la Côte d'Azur

📅 2018 – 2021

📍 Nice, France

Context: Engineering and analysis activities for gravitational-wave mission preparation, involving signal analysis, simulation, software workflows and noise characterization.

Objective: Develop robust analysis methods and validation workflows for complex signals and demanding system environments.

- **Simulation workflows:** Built simulation approaches for complex signal environments, uncertainty studies and large-scale datasets.
- **Software validation:** Compared simulated outputs, model behaviours and analysis results to assess consistency and robustness.
- **Technical investigations:** Investigated noise sources through cross-sensor correlation analyses in diagnostic contexts.
- **Performance analysis:** Supported the identification of limiting factors affecting mission-level performance and system sensitivity.
- **Scientific computing:** Developed strong expertise in Python-based analysis, modelling, documentation and reproducible workflows.

Selective Graduate Program in Fundamental Physics (Magistère)

Université Paris-Saclay

📅 2016 – 2018

📍 Orsay, France

MSc in Astronomy and Astrophysics

Observatoire de Paris / Université Paris-Saclay

📅 2017 – 2018

📍 Paris, France

BSc in Fundamental Physics

Université Paris-Saclay

📅 2015 – 2016

📍 Orsay, France

PROFESSIONAL TRAINING

Supervised Machine Learning (Regression & Classification)

DeepLearning.AI - Andrew Ng

📅 2020

📍 Coursera

Recherche reproductible : principes méthodologiques pour une science transparente

FUN MOOC – Inria

📅 2025

📍 France Université Numérique

Continuous Professional Training – Software, Data, AI and Systems

OpenClassrooms

📅 2024 – 2026

Python, SQL, Machine Learning, AI, Linux, JavaScript, TypeScript, Angular, Node.js, Django, REST APIs, C/C++, web development, algorithms and software engineering fundamentals.

PROJECTS

CNES / LISA – Software/System Coordination, Requirements Logic & Validation

ESA/NASA Space Mission - Large-scale International Collaboration

📅 2023 – 2025

📍 Nice, France

Coordination and structuring of software, data-processing and validation workflows for the LISA space mission within a large international engineering and scientific collaboration.

Project management relevance: Cadrage of technical activities, contributor coordination, validation planning, delivery follow-up, risk identification, reporting, documentation and stakeholder alignment.

Software/system relevance: Work involving complex software workflows, data-processing chains, GitLab-based development, reproducible environments, technical documentation, validation evidence and versioned deliverables.

Scale: Major international space mission with an estimated budget of approximately 2 billion EUR over 10 years.

Einstein Telescope / LIGO–Virgo–KAGRA – Technical Studies & Collaborative Project Workflows

International Collaborations

📅 2021 – 2023

📍 Antwerp, Belgium

Contribution to collaborative developments in data analysis, software methods and technical investigations within large-scale international scientific and engineering collaborations.

Project management relevance: Coordination of technical studies, reporting, documentation, contribution to collaborative review, prioritisation of analysis tasks and communication with international stakeholders.

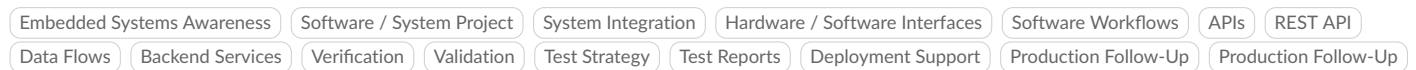
System impact: Studies on environmental and instrumental noise sources supporting the identification of performance limitations and design constraints for next-generation gravitational-wave observatories.

STRENGTHS

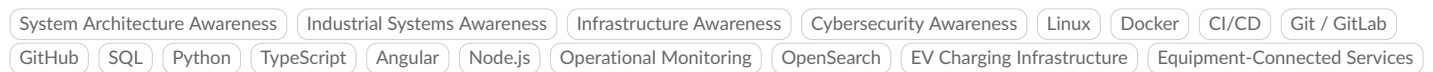
Project management & delivery

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|--------------------|---------------------------|-----------------------------|--------------------|----------------------------|---------------------|--------------------|----------------|
| Project Management | Project Coordination | Cadrage | Needs Analysis | Requirements Formalisation | Functional Analysis | Technical Analysis | Specifications |
| Cahier des Charges | Requirements Traceability | Architecture Review Support | Planning Follow-Up | Risk Identification | Action Plans | Delivery Support | |
| Quality Follow-Up | Cost Awareness | Decision Support | | | | | |

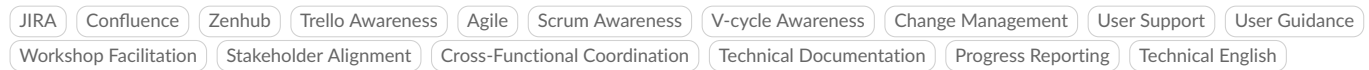
Software/system integration, verification & validation



Architecture, cloud & technical environment



Tools, methods & change management



Office & communication

